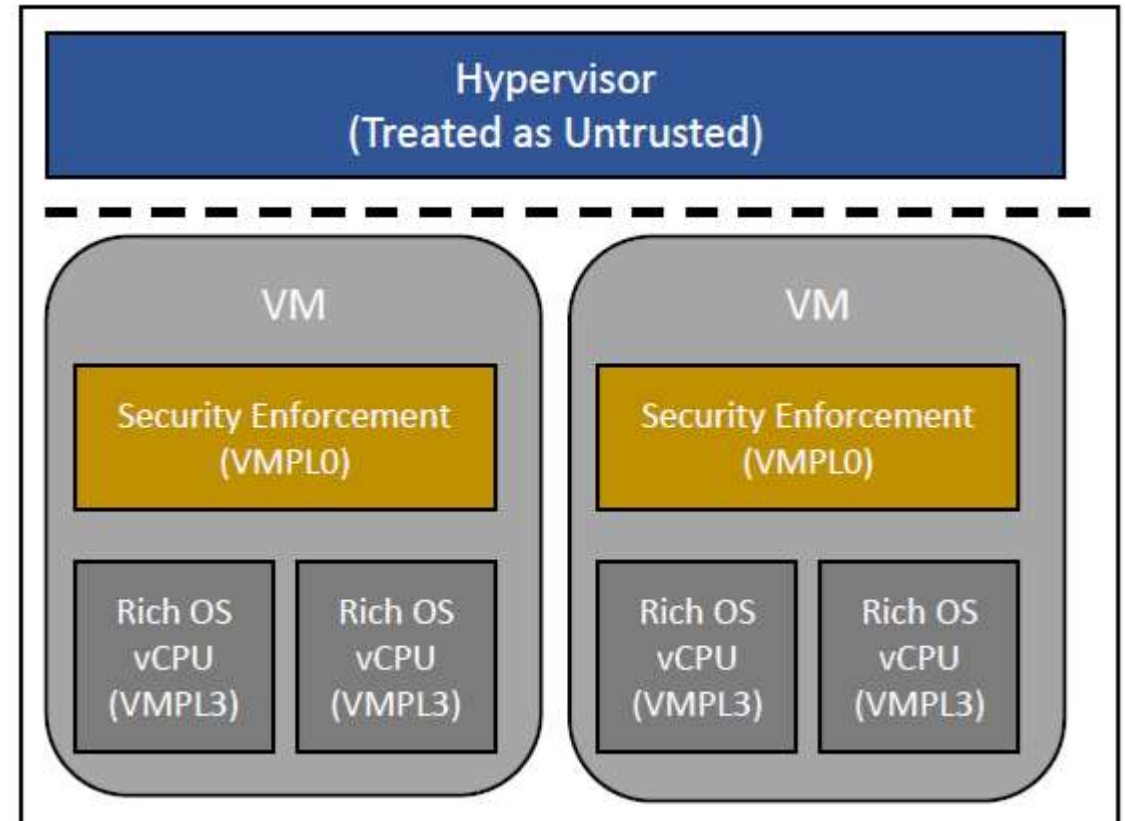


Virtual Machine Privilege Levels (VMPLs)

- VMPLs are an optional feature allowing guest VMs to divide their address space into four hierarchical levels.
- VMPL0 is highest privilege, VMPL3 is least privileged.
- This enables hardware-isolated abstraction layers within a VM for additional security controls.



VMPL Use Cases

Secure Hypervisor Virtualization

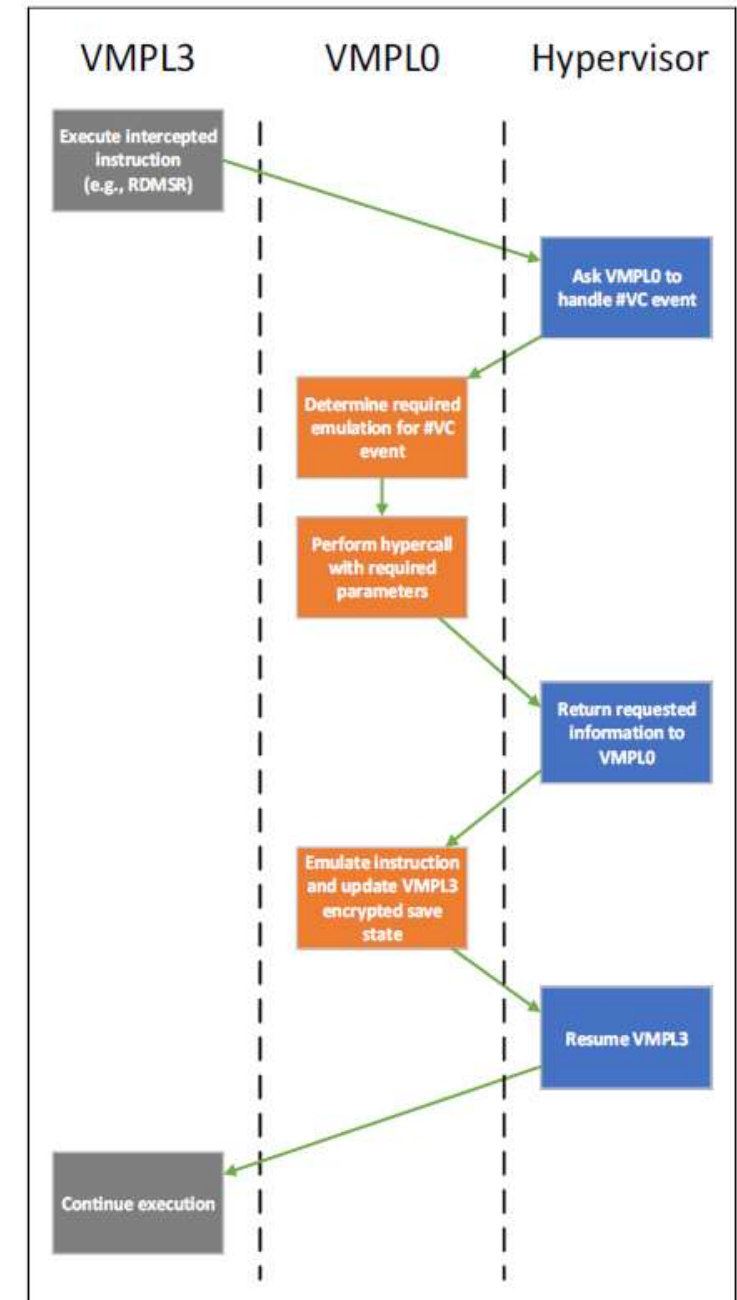
VMPL0 enforces page permissions like a bare-metal hypervisor would, enabling secure virtualization of security-enforcing hypervisors in untrusted cloud environments.

Trusted APIC Emulation

VMPL0 performs trusted APIC emulation inside guest trust domain, while rest of guest runs in lower VMPLs unaware of emulation.

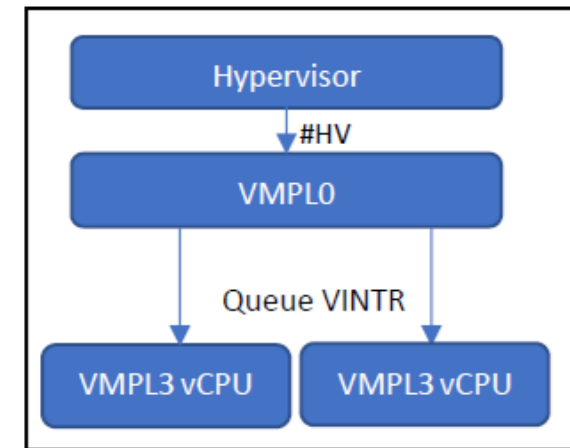
Legacy Guest Support

VMPL0 acts as intermediary for guest-hypervisor communications, handling #VC exceptions and C-bit management, enabling unenlightened legacy VMs to use SEV-SNP.



Interrupt and Exception Protection

- SEV-SNP adds two optional modes for more restrictive interrupt/exception interfaces between VM and hypervisor, protecting against violations of OS assumptions about interrupt behavior.
- Restricted Injection
 - Disables virtual interrupt queuing and partially disables interrupt injection interface.
 - Hypervisor can only inject new #HV exception vector as doorbell.
 - VM and hypervisor communicate via para-virtualized event queue in shared memory.
- Alternate Injection
 - Allows standard virtual interrupt queuing and injection, but only controlled by guest itself.
 - New fields in encrypted Virtual Machine Save Area (VMSA) allow interrupt queuing and event injection.
 - All interrupt-related security-sensitive state saved to VMSA.

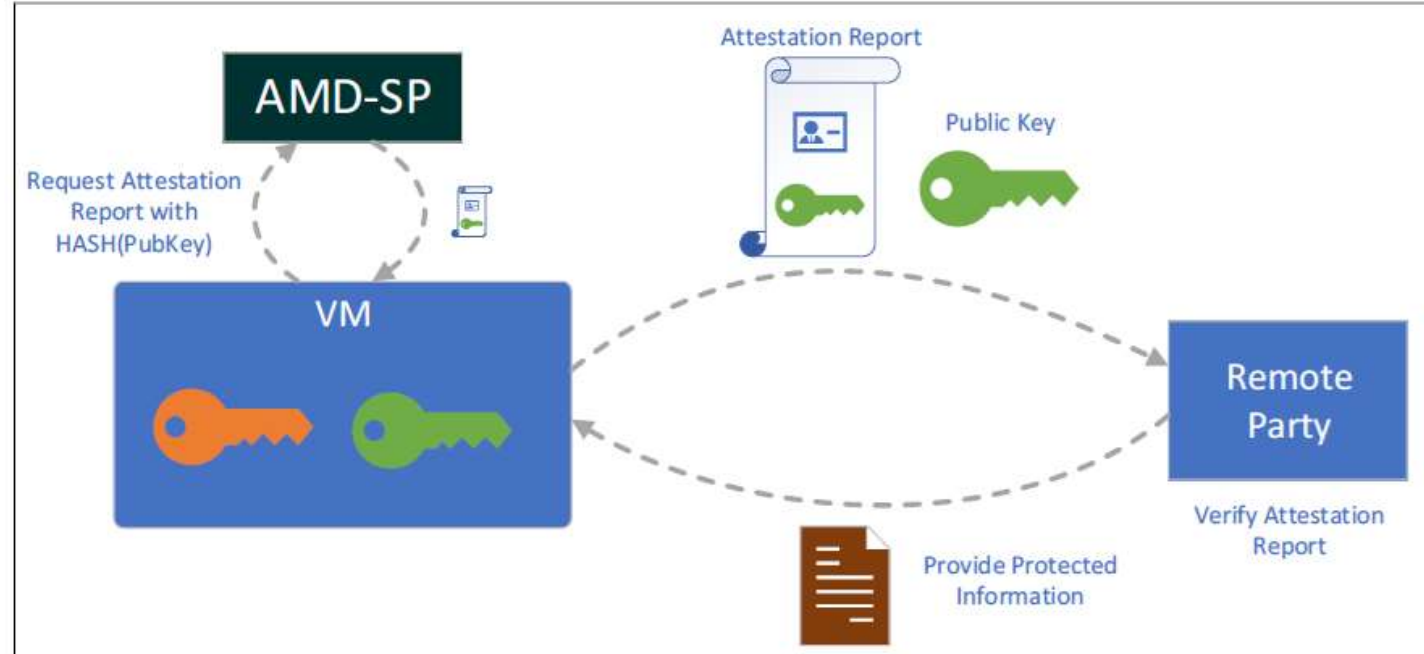


VM Launch Process

■ Process

- SEV-SNP VMs start from initial unencrypted image containing boot code but no secrets.
 - Hypervisor asks AMD-SP (AMD Secure Processor) to install initial pages.
 - AMD-SP cryptographically measures contents and metadata into launch digest.
- ## ■ Guest owner supplies signed Identity Block (IDB) associating with VM.
- IDB contains fields to uniquely identify VM and expected launch digest.
 - Only VMs matching expected digest can be associated with IDB.

Attestation Process



- Unlike SEV/SEV-ES, SEV-SNP supports attestation at any time through protected path

VM Migration

SEV-SNP enhances migration flexibility by offloading policy enforcement to a Migration Agent (MA)—itself an SEV-SNP VM running on same physical system as primary VM. Each VM can be associated with one MA, but single MA can manage migration for arbitrary number of VMs.

Complex Policy Implementation

MA determines what systems primary VM can migrate to, implementing complex policies using any desired means. MA information included in VM attestation report as part of TCB.

Flexible Migration Timing

Source and destination machines don't need to be online simultaneously. MA can hold VM state or put into long-term storage, reconstituting VM later on same or different machine.

Secure Transfer Process

MA on source machine authenticates MA on destination machine and establishes protected network connection. If successful, transfers required guest information for reconstitution.